

Model Report: Credit Risk Modelling Platform

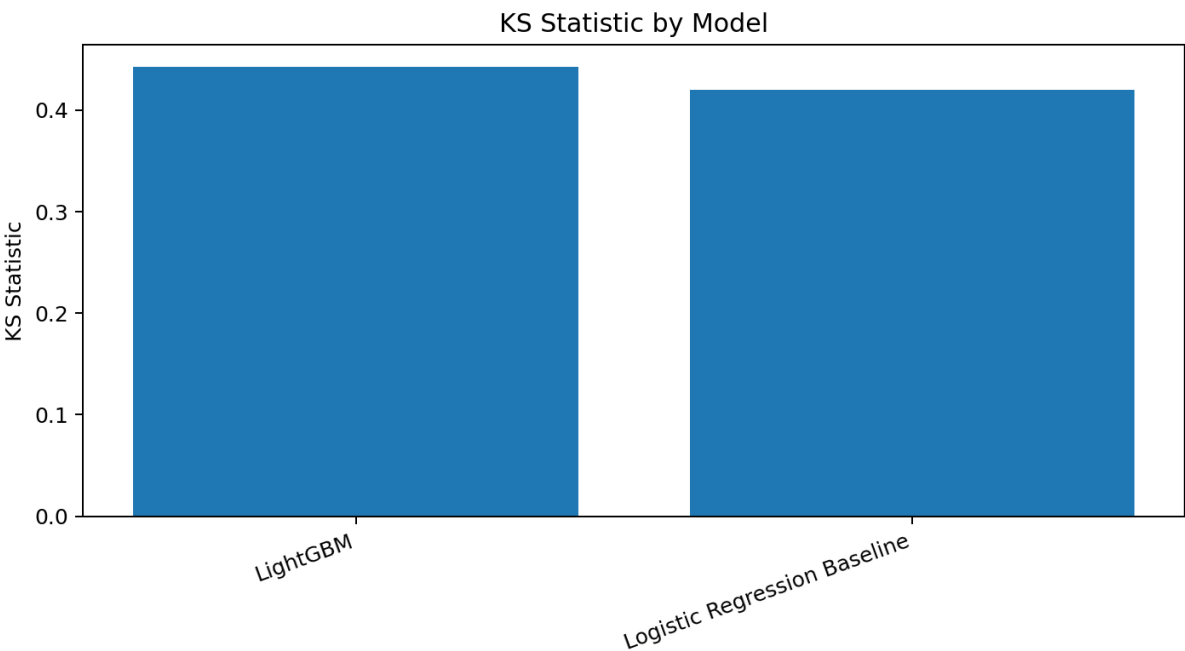
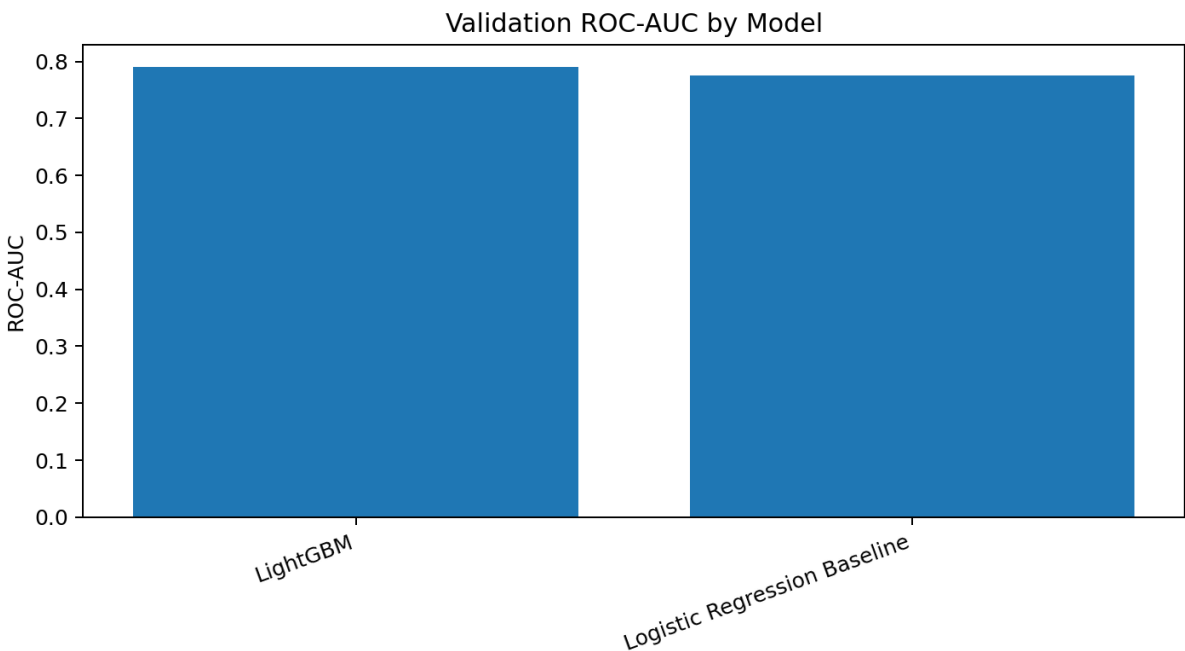
This report summarises the model training, validation, ranking performance and explainability outputs for the Home Credit default-risk project.

Executive Summary

The selected LightGBM model achieved a validation ROC-AUC of **0.7900**, compared with **0.7755** for the Logistic Regression baseline. Its KS statistic was **0.4423**, showing stronger separation between higher-risk and lower-risk applicants.

Model Comparison

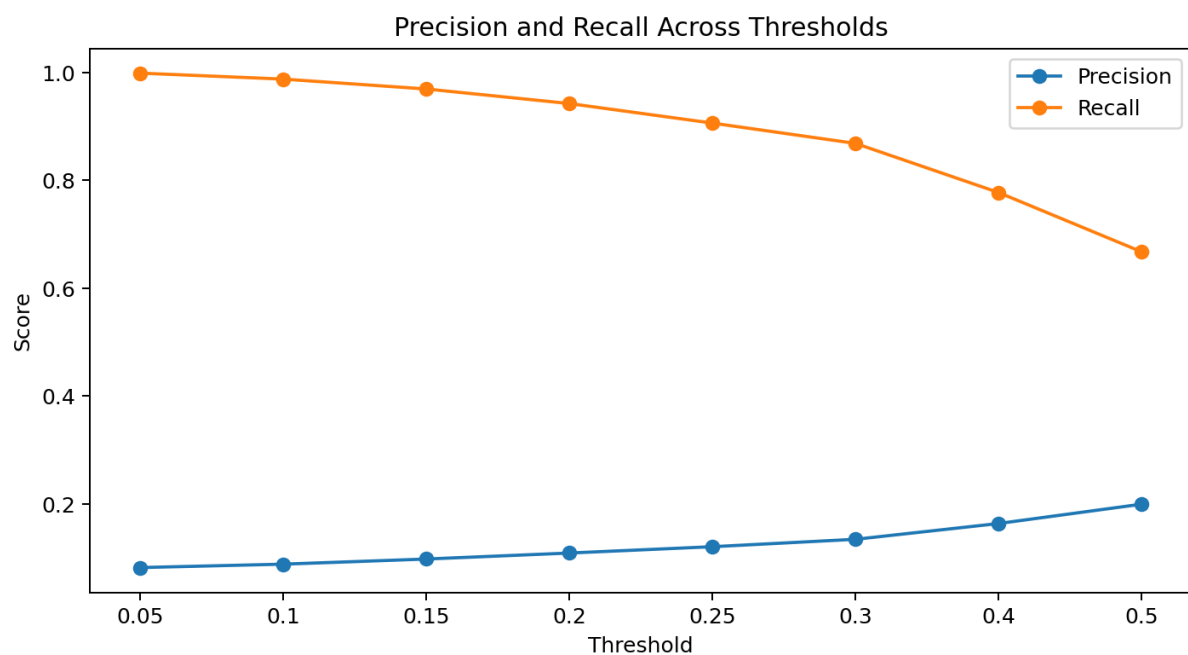
model_name	roc_auc	gini	ks_statistic	accuracy	precision	recall	f1_score
LightGBM	0.7900	0.5800	0.4423	75.72%	19.97%	66.75%	0.3074
Logistic Regression Baseline	0.7755	0.5511	0.4201	71.16%	17.75%	70.80%	0.2839



Threshold Performance

Lower probability thresholds identify more true defaults but increase false positives. Higher thresholds are more selective but miss more default cases.

threshold	accuracy	precision	recall	f1_score	false_positives	false_negatives	true_positives
0.05	9.95%	8.22%	99.88%	0.1519	55378.0000	6.0000	4959.0000
0.10	17.77%	8.85%	98.77%	0.1624	50512.0000	61.0000	4904.0000
0.15	27.73%	9.80%	96.94%	0.1780	44298.0000	152.0000	4813.0000
0.20	37.42%	10.91%	94.24%	0.1956	38200.0000	286.0000	4679.0000
0.25	46.04%	12.09%	90.61%	0.2133	32723.0000	466.0000	4499.0000
0.30	53.87%	13.46%	86.85%	0.2331	27721.0000	653.0000	4312.0000
0.40	66.18%	16.39%	77.74%	0.2707	19698.0000	1105.0000	3860.0000
0.50	75.72%	19.97%	66.75%	0.3074	13282.0000	1651.0000	3314.0000



Decile and Lift Analysis

The highest-risk decile captured **38.45%** of all defaults while representing **10.00%** of applicants. This demonstrates strong risk concentration in the top-scored population.

risk_decile	applicant_count	default_count	actual_default_rate	default_share	cumulative_default_share	lift
9.0000	6151.0000	1909.0000	31.04%	38.45%	38.45%	3.84
8.0000	6150.0000	949.0000	15.43%	19.11%	57.56%	1.91
7.0000	6150.0000	636.0000	10.34%	12.81%	70.37%	1.28
6.0000	6150.0000	425.0000	6.91%	8.56%	78.93%	0.86
5.0000	6150.0000	336.0000	5.46%	6.77%	85.70%	0.68
4.0000	6151.0000	229.0000	3.72%	4.61%	90.31%	0.46
3.0000	6150.0000	201.0000	3.27%	4.05%	94.36%	0.40
2.0000	6150.0000	129.0000	2.10%	2.60%	96.96%	0.26
1.0000	6150.0000	88.0000	1.43%	1.77%	98.73%	0.18
0.0000	6151.0000	63.0000	1.02%	1.27%	100.00%	0.13

Model Explainability

metric	value
model_file	/Users/shriyanshnautiyal/Downloads/credit-risk-modelling-home-credit/models/credit_risk_model.pkl
feature_count	618
shap_sample_size	5000
top_feature_by_model_importance	APP_ANNUITY_TO_CREDIT_RATIO
top_feature_by_shap_importance	EXT_SOURCE_2

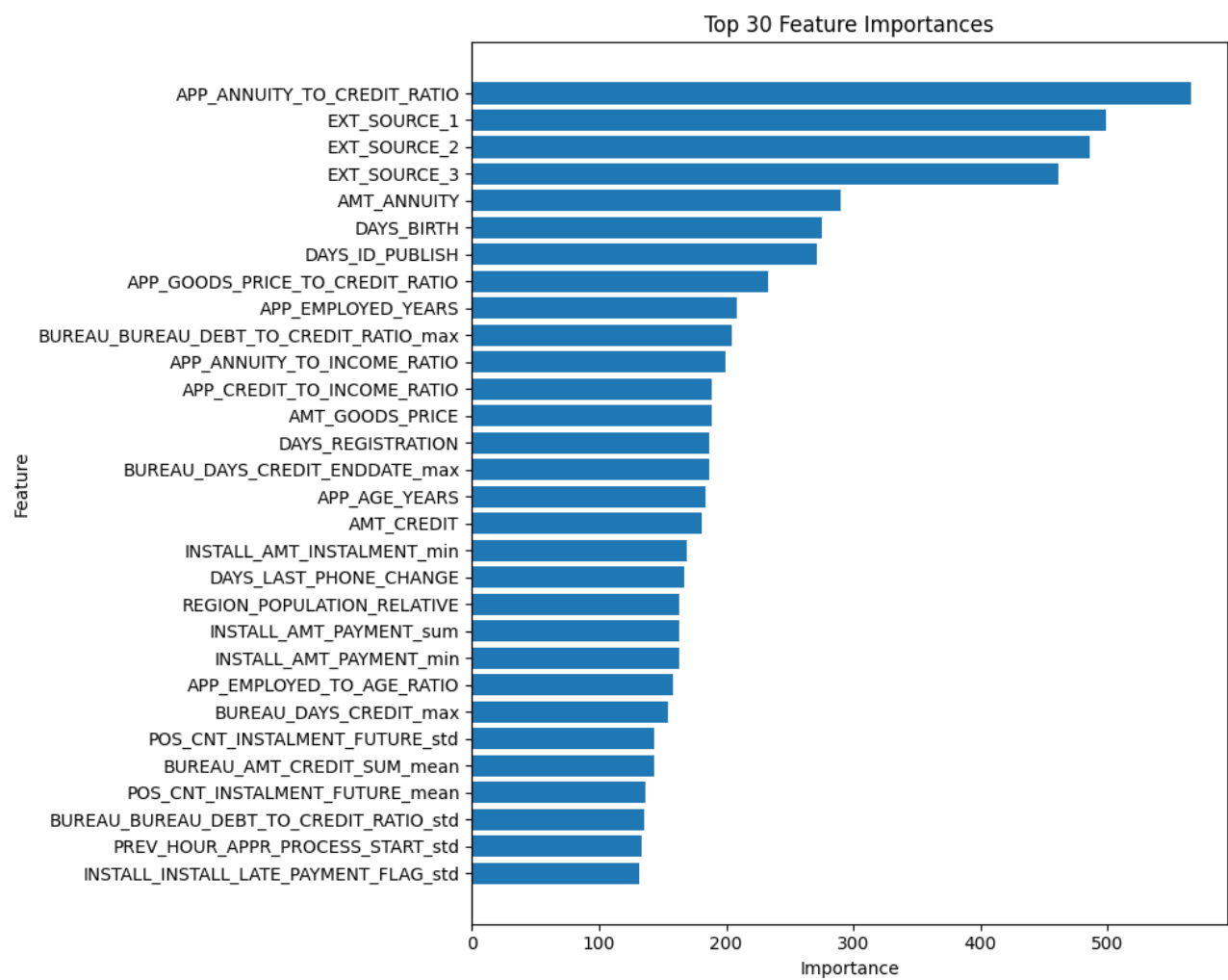
Top SHAP Features

feature	mean_absolute_shap_value
EXT_SOURCE_2	0.3231
EXT_SOURCE_3	0.2593
EXT_SOURCE_1	0.1464
APP_GOODS_PRICE_TO_CREDIT_RATIO	0.0999
AMT_ANNUITY	0.0947
BUREAU_BUREAU_DEBT_TO_CREDIT_RATIO_max	0.0897
APP_ANNUITY_TO_CREDIT_RATIO	0.0874
APP_EMPLOYED_YEARS	0.0840
NAME_FAMILY_STATUS_Married	0.0659
CODE_GENDER_F	0.0633

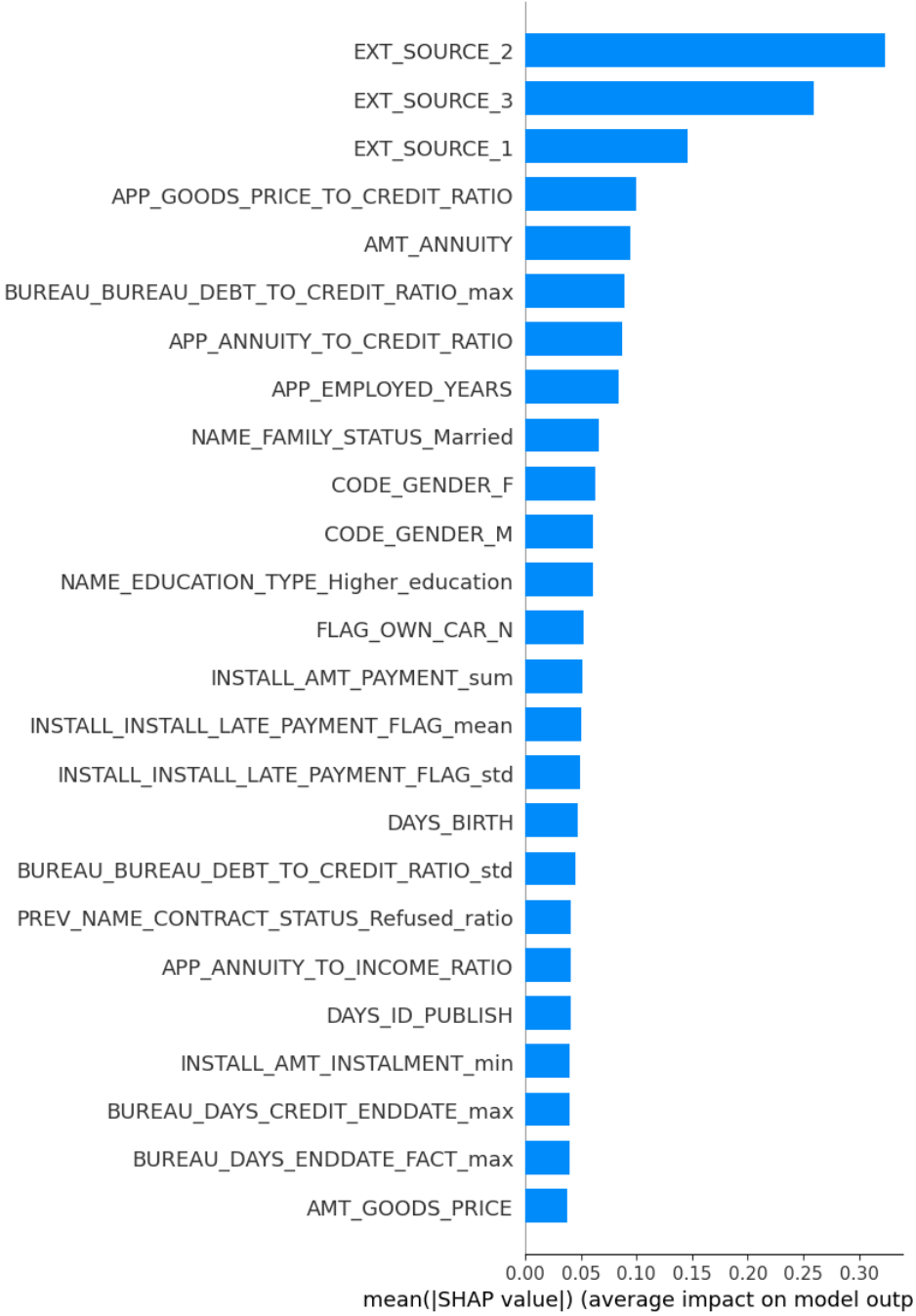
Top LightGBM Features

feature	importance
APP_ANNUITY_TO_CREDIT_RATIO	566
EXT_SOURCE_1	499
EXT_SOURCE_2	486
EXT_SOURCE_3	461
AMT_ANNUITY	290
DAYS_BIRTH	275
DAYS_ID_PUBLISH	271
APP_GOODS_PRICE_TO_CREDIT_RATIO	233
APP_EMPLOYED_YEARS	208
BUREAU_BUREAU_DEBT_TO_CREDIT_RATIO_max	204

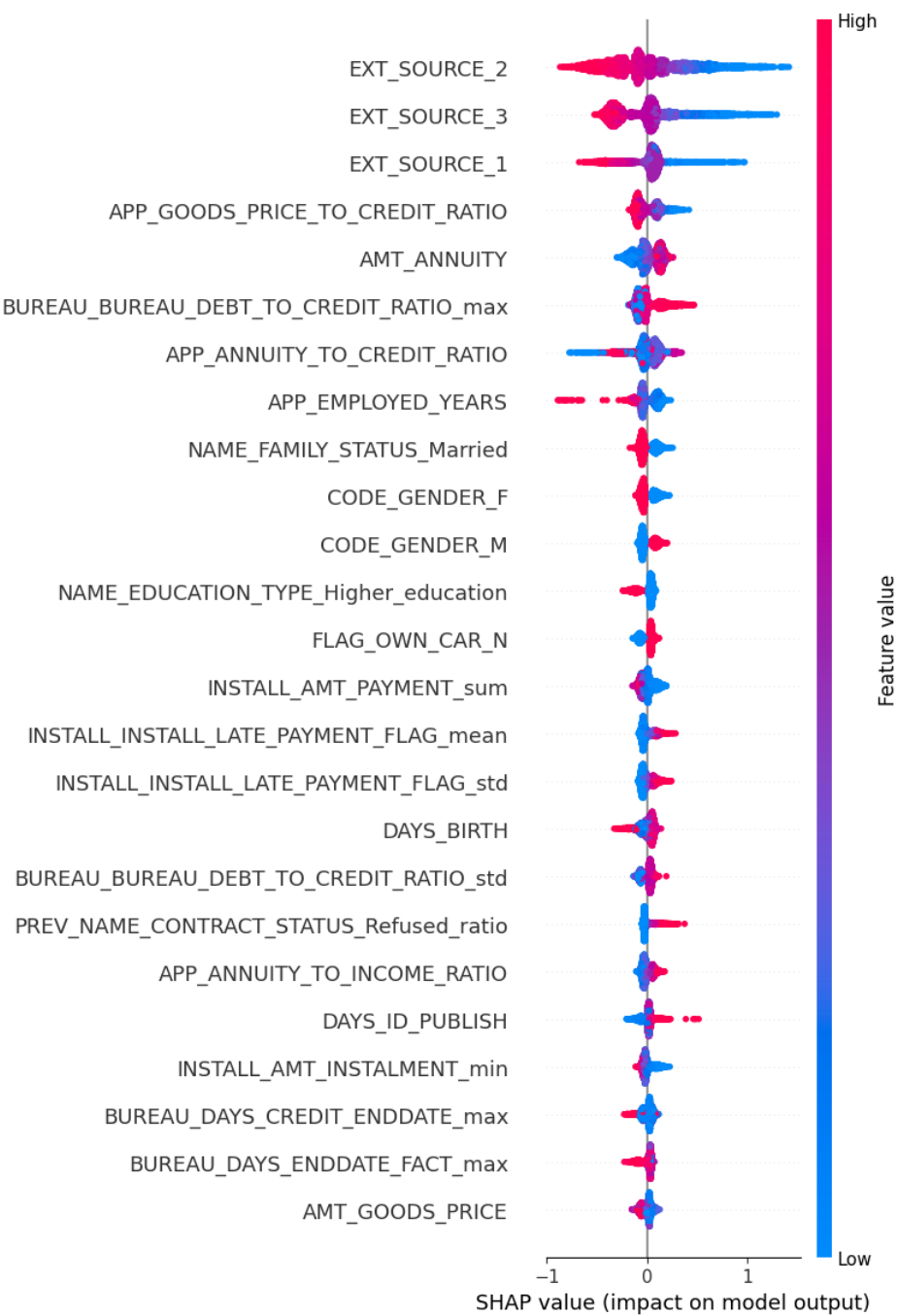
Feature Importance Plot



SHAP Bar Plot



SHAP Summary Plot



Limitations

The model is built on a public competition dataset and uses illustrative modelling choices. A production credit-risk model would require calibration, fairness analysis, policy-rule integration, regulatory review, live monitoring and governance approval.